

```

doc 1.1.0
evolution 1.1.0
iscsid 1.0.0
mozilla 1.1.0
mplayer 1.1.0
nagios 1.1.0
odkjob 1.0.1
openoff 1.0.0
pcscd 1.0.0
pyzor 1.1.0
razor 1.1.0
ricci 1.0.0
smartmon 1.1.0
tftp 1.0.1
vbg 1.0.3

```

What this means is that the default SELinux installation does come with some modules loaded that are not part of the base policy. Looking more closely into the output of the above command, we see that in my system there are 18 policy modules installed. Each row of the above output corresponds to a policy module.

The output of the `semodule -l` command gives us two columns of information: *Module Name* and *Module Version*. Thus we can see that the *amavis* module has a version number of 1.1.0 whereas the *vbg* module has a version 1.0.3.

Also, these are the modules currently loaded into the memory and are active along with the base policy. But, where are these modules located? What difference do they make to the overall SELinux policy? How are they loaded and removed?

Let's try to answer the above questions, one by one.

These are binary policy modules that, by default, have a file extension of *.pp* (Policy Package). Generally, the module name and the file name is kept the same, though it's not mandatory. Therefore, we need to look up a file named *amavis.pp* (Policy Package for the Amavis daemon). By default, the location of this file is `/etc/selinux/targeted/modules/active/modules/`. If you go into this folder and list the contents, you will see the policy package files of all the currently loaded modules.

The policy modules obviously make a lot of difference. That's what they were created for. To observe the difference they make, let's use the wonderful *seinfo* tool discussed earlier.

To get an overall idea of the SELinux policy currently loaded, we shall use the command *seinfo*. A sample output from my system is shown below:

```
[root@vbg modules]# seinfo
```

```

Statistics for policy file: /etc/selinux/targeted/
policy/policy.21
Policy Version & Type: v.21 (binary, MLS)

```

```

Classes:      61  Permissions:  220
Types:       1516 Attributes: 148
Users:        3   Roles:    6
Booleans:    211  Cond. Expr.: 187
Sensitivities: 1 Categories: 1024
Allow:       82576 Neverallow: 0
Auditallow:  28  Dontaudit: 9086
Role allow:   5   Role trans: 0
Type_trans:  1400 Type_change: 17
Type_member:  0   Range_trans: 23
Constraints:  47  Validatetrans: 0
Fs_use:       15  Genfscon: 64
Portcon:     264  Netifcon: 0
Nodecon:      8   Initial SIDs: 27

```

We can see that there are 1,516 types and 82,576 allow rules being recognised by SELinux. You can redirect this output to a temporary file just for comparison, later. You could use the following command:

```
[root@vbg modules]# seinfo > /tmp/org-selinux-policy
```

Let's now remove one of the loaded modules. As an example, let us remove the *amavis* module.

Removing SELinux modules

To remove a loaded SELinux module, use the *semodule* command with the *r* option and the module name as the argument. For example:

```
[root@vbg modules]# semodule -r amavis
```

This removes the *amavis* module. To ensure that the above command

FOSTER Linux
File & Open Source Training Education & Research

India's ONLY
 OpenSource training
 institute by
**EXPERIENCED System
 Administrator's &
 Solution Architects**

25% Off on
 Select
 trainings

The November Training Calendar

- **Fast Track RHCE:**
 4th Nov 09 & 11th Nov 09
- **Regular RHCE:**
 9th Nov 09 (7am – 9am)
 9th Nov 09 (6pm – 9pm)
- **RH 423 LDAP:**
 5th to 8th Nov 09
- **RH 429 SELinux**
 11th to 14th Nov 09
- **Clustering & Storage**
 20th to 22nd Nov 09
- **PostgreSQL**
 20th to 22nd Nov 09
- **PHP & MySQL**
 18th Nov 09
- **Shell Scripting**
 30th Nov 09
- **Mail Server**
 9th - 11th Nov 09
- **JBoss**
 14th Nov 09 (Sat & Sun)

RHCE @
 Rs 10,500
 ONLY

Exam Dates
 • **RHCE:**
 8th & 25th Nov 09
 • **EX 423:**
 16th Nov 09
 • **EX 429:**
 16th Nov 09
 • **EX 333:**
 17th Nov 09

SCO 45-37, 8 Floor
 Old GUP Colony, Sector - 14
 Gurgaon - 122 001, Haryana
 Tel: +91-124-4268167 / 4090880
 Email: info@fosteringlinux.com
 Website: www.fosteringlinux.com

redhat
 TRAINING PARTNER